Reg. No. [ ] [18AUDE761]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

OFF ROAD VEHICLES

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List the common important characteristics of the off road vehicles.	**04**
	b.	What is prime mover? Classify the prime movers. With a neat sketch explain the Open and closed gas turbines.	**10**
2.	**a.**	List the factors required to improve the ripping efficiency.	**04**
	b.	Explain the following with neat sketch: **i.** Bucket wheel excavator **ii.** Power shovel	**10**
3.	**a.**	What is Jib Crane? With neat sketch explain the working principle of the Jib crane.	**10**
	b.	List the industrial applications of the Fork lift.	**04**
		OR	
4.	**a.**	What is soil compaction? List the reason for soil compaction.	**04**
	b.	Briefly explain the purpose of generators at construction sites? Explain the diesel engine power plant with a neat sketch.	**10**
5.	**a.**	Explain any two farm machineries used for soil cultivation & fertilizing and pest control.	**04**
	b.	Classify the farm tractor. With a neat sketch, explain any six important components of the tractor.	**10**
		OR	
6.	**a.**	With a flow chart and neat diagram explain the combined Paddy harvester.	**10**
	b.	List any four merits and limitations of the farm mechanization.	**04**
7.	**a.**	Give the detailed classification of the brake. Explain the hydraulic brake with a net sketch.	**06**
	b.	A caliper brake is required for the front wheels of a passenger car with a braking capacity of 320 Nm for each brake. Preliminary design estimates have set the brake geometry as r_i = 100 mm, r_o = 140 mm, and θ = 40°. Pads with a coefficient of friction of 0.35 have been selected. Each pad is actuated by means of a hydraulic cylinder of nominal diameter 25.4 mm. Determine the required actuating force, the average and the maximum contact pressures, and the required hydraulic pressure for brake actuation.	**08**
		OR	
8.	**a.**	With a neat block diagram explain the basic components of the pneumatic system.	**08**
	b.	With a neat diagram explain the working principle of hydraulic brake.	**06**